SEGi University

Faculty of Engineering, Built Environment and Information Technology

BCL2113 Software Engineering

Continuous Assessment: 40%

Applying Software Engineering Methodologies in LMS Development

A Camu informed proposal for a private university

Student name: Chan Jing Yi

Student ID: SUOL2500321

Report date: 5 November 2026

Main report: 2,022 words
Including headings, tables and captions; excluding diagram labels, references, front matter and Appendix A.

Table of contents

1. Planning and analysis 1
2. Requirements engineering 1
Functional requirements and analysis 2
Nonfunctional requirements and validation 3
3. Scrum process model 4
4. System design 4
5. Team roles and project planning 7
6. Software quality and testing 8
7. Risk management 9
8. Ethical and legal considerations 10
References 11
Appendix A. Real user elicitation instrument 12

List of figures

Figure 1. Proposed LMS use case diagram 5
Figure 2. Assignment submission domain classes 6
Figure 3. Likelihood and impact risk matrix 9

List of tables

Table 1. Functional requirements and acceptance conditions 2
Table 2. Nonfunctional requirements and verification 3
Table 3. Proposed lifecycle timeline 7
Table 4. Submission test cases and expected outcomes 8
Table 5. Risk responses and owners 9

1. Planning and analysis

1.1 Proposed system and evidence

The proposed University LMS improves course delivery for a mid sized private university, using SEGi's Camu experience as its case context. Its scope covers enrolment, materials, assignments, grades and course communication. Attendance and finance remain outside the initial release.

A public MyCamu review reports unsuccessful downloads and unexpected logout (Kiruba, 2026). An older review reports submission and timetable notification difficulties (Suriyakumar, 2021). These individual accounts identify concerns, but cannot establish their frequency or current occurrence at SEGi. Conversely, university guidance documents assignment deadlines and completion status within MyCamu (Punay, 2026). The proposal therefore improves discoverability and verification without assuming these capabilities are absent.

1.2 Objectives

Objective one is to provide enrolled students with a unified deadline dashboard and consistent course workspace before the next academic year. Objective two is to make assignment submission verifiable and protect academic information through receipts, permissions and tested recovery procedures. Proposed acceptance targets appear in the specification.

2. Requirements engineering

2.1 Elicitation and evidence status

On 4 November 2026, four students provided anonymised feedback about their own Camu accounts, with agreement to use their responses in this report. They rated ease of finding materials and deadlines as 2, 3, 2 and 3 out of 5. Their responses are summarised in Appendix A. The small student sample does not represent all users; lecturer and administrator validation remains necessary.

All four described a mismatch between Tasks and the empty Assignments view and reported a Messages error. Several wanted course materials in Content, clear deadlines and links from tasks to feedback. The calendar and task status labels were valued. Empty views do not establish a defect: materials may be unpublished or available through a course page. A later course-specific check found Software Engineering materials and its assessment in Content. Login, downloads and mobile use were not verified.

2.2 Functional user requirements

Students access enrolled courses; lecturers manage assigned courses; administrators manage membership and staff assignments. All requirements are provisional until users validate them.

Table 1. Functional requirements and acceptance conditions

ID	Required system behaviour
FR1	Allow an authenticated student to enrol in an open course. Reject duplicate membership and show confirmation.
FR2	Let assigned lecturers publish materials in a standard course area. Only enrolled students and authorised staff can download them.
FR3	Accept a student's PDF or DOCX assignment up to 20 MB. Issue a unique receipt only after successful storage and record creation.
FR4	Let lecturers create assignments and publish deadlines, grades and feedback. Students see only their own private results.
FR5	Provide course announcements and messages to enrolled students and assigned lecturers, with a visible delivery status.
FR6	Display published deadlines chronologically; show the same deadline and submission status in every assignment view; allow calendar export.
FR7	Let students reopen receipts showing filename and server receipt time, and download the exact stored submission.
FR8	Let administrators create courses and assign lecturers and memberships. Reject unauthorised administrative requests.

2.3 Analysis and traceability

The student responses support five needs: accessible course materials (FR2), consistent assignment information across views (FR4 and FR6), visible deadlines (FR6), reliable messages (FR5), and accessible grades and feedback (FR4). Student 2 requested feedback from the task; Student 4 wanted helpful empty states. Students 3 and 4 valued the calendar, while Students 1 and 2 highlighted task statuses. Proposed receipts (FR3 and FR7) address submission assurance but were not established by these responses.

Public reviews are secondary evidence and may concern other versions or institutions. The historical 2021 review supports investigation, not a claim about current SEGi behaviour. The Product Owner prioritises receipts over calendar export because lost submission evidence has greater consequences, then seeks student and lecturer review. Requirements and targets require stakeholder validation.

2.4 Nonfunctional requirements

The following values are proposed acceptance targets, not observed Camu performance. Validate the workload and service commitments with university stakeholders.

Table 2. Nonfunctional requirements and verification

ID	Measurable requirement
NFR1	Deny every tested attempt to read another student's private submission or grade without an authorised staff role.
NFR2	Under 500 active simulated sessions, complete 95% of course page requests within three seconds, excluding file transfer.
NFR3	Achieve 99.5% monthly availability excluding announced maintenance, measured by external monitoring.
NFR4	Restore encrypted daily backups within four hours, losing at most 24 hours of records, demonstrated in a recovery exercise.
NFR5	Meet WCAG 2.2 AA across assessed student journeys, verified through automated checks and manual keyboard and screen reader testing.
NFR6	Reject disallowed, oversized or malicious files before release. Validate content and permissions and store uploads outside public executable paths.

2.5 Requirements validation and control

Elicitation is followed by analysis, specification, validation and controlled revision. Replace vague statements such as "fast uploads" with measurable conditions. Check contradictions through shared definitions: a deadline uses server time, a submission completes when storage and metadata succeed, and late work requires an authorised extension. Failed transfers produce no success receipt.

Each requirement links to its source, owner and acceptance test. Stakeholders review exception scenarios before approval; a change request records impacts on cost, tests and schedule. For example, a new file type needs Product Owner approval, security review, implementation and verification before closure. Accessibility follows W3C (2024), while upload controls follow OWASP (n.d.). Neither guidance replaces local user validation.

For mobile sessions, preserve draft form content before expiry, warn users before requiring authentication and restore the draft afterwards. Do not extend access indefinitely or bypass permission checks. Confirm the university's preferred timeout policy with administrators. Test interruptions by switching applications and reconnecting after network loss; the expected behaviour must distinguish a saved draft from an accepted assignment.

3. Scrum process model

3.1 Implementation and suitability

Scrum provides short feedback cycles for uncertain requirements (Schwaber and Sutherland, 2020). The Product Owner orders the Product Backlog around a reliable course workspace. Two week Sprints produce usable increments, beginning with enrolment and progressing through materials, submissions and grading.

Sprint Planning establishes a Sprint Goal and selected work; developers adapt their plan during the Daily Scrum. In a submission Sprint, validation, storage and receipts form one usable slice. At the Sprint Review, students and lecturers inspect it and influence backlog priorities. The Retrospective improves working practices, such as preventing repeated upload defects.

The Definition of Done requires reviewed code, passing acceptance tests, accessibility checks and updated documentation. Scrum suits the LMS because feedback can reveal policy exceptions before release. It cannot guarantee a fixed scope and launch date; lower priority work must remain negotiable.

4. System design

4.1 Modelling approach

The following diagrams describe the proposed system, not Camu's internal implementation. Figure 1 maps actors to all eight functional requirements. Figure 2 models assignment submission with attributes, operations and multiplicities. Authentication is a precondition; membership determines authorisation. In MVC, the model stores submission data, the view displays the form and receipt, and the controller handles requests.

4.2 Use case diagram

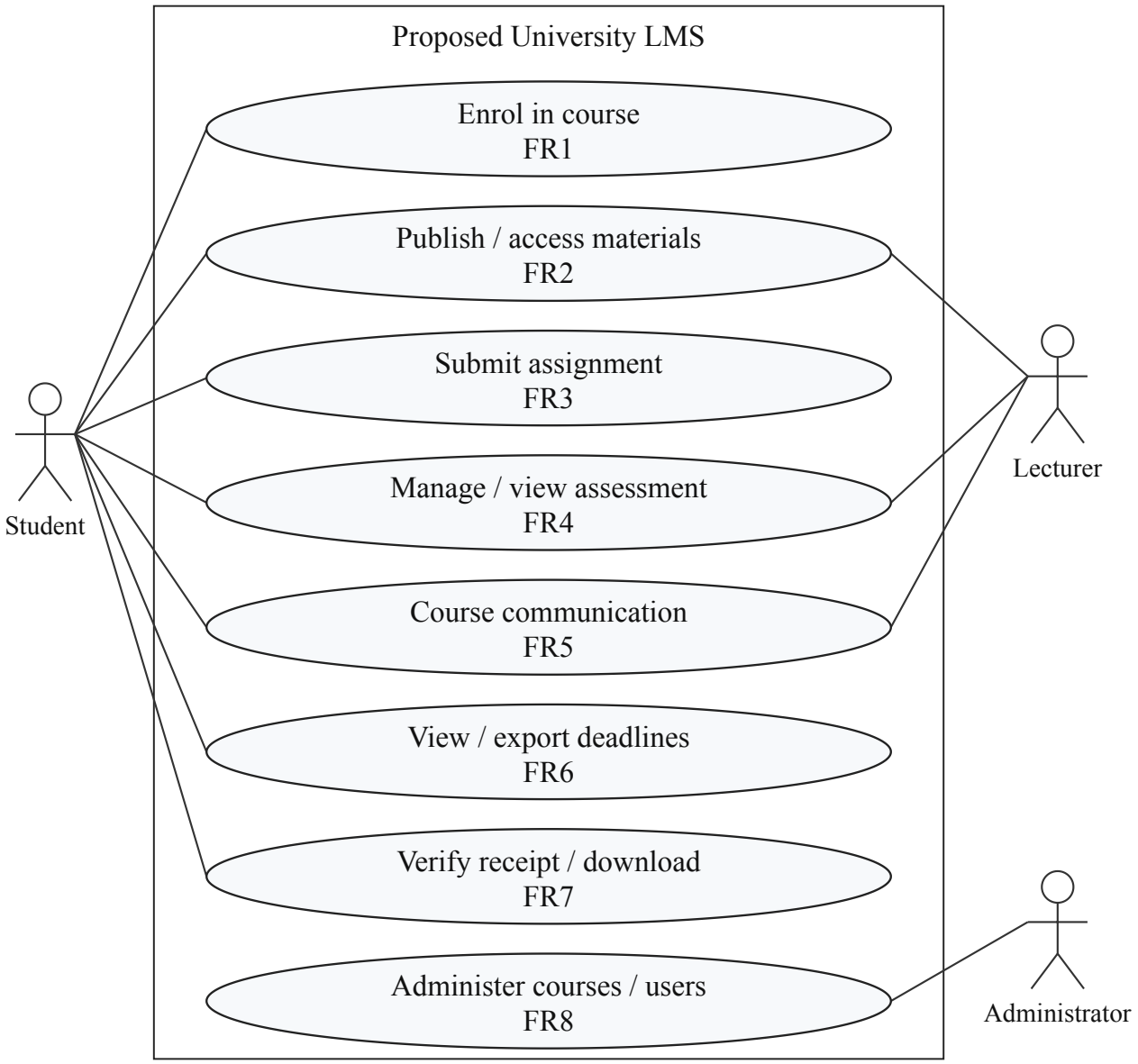


Figure 1. Proposed LMS use case diagram

Student associations cover enrolment, materials, submissions, communication, deadlines and receipts. Lecturers publish content, assess work and communicate. Administrators manage course access. Solid associations indicate participation rather than execution order. FR4 groups assignment management and assessment; permission checks distinguish lecturer actions from student result viewing.

4.3 Assignment submission class diagram

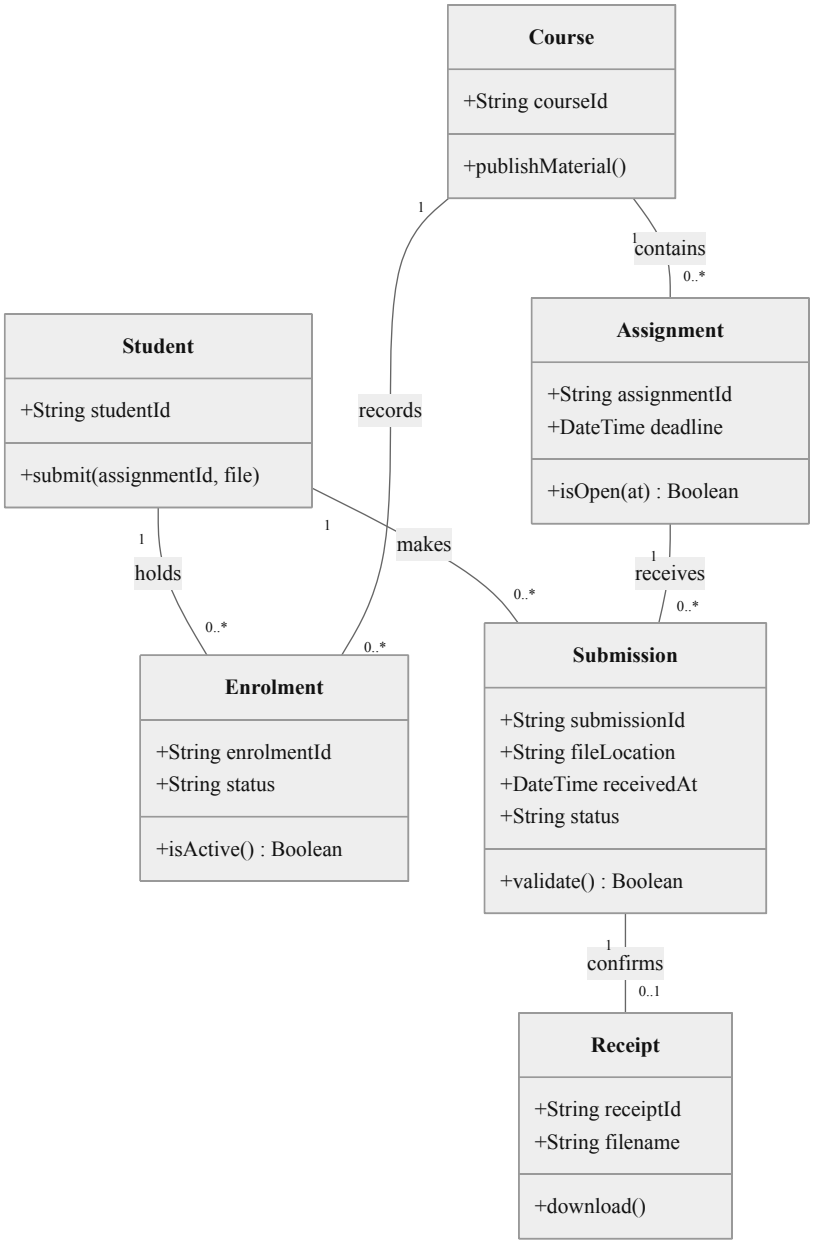


Figure 2. Assignment submission domain classes

Enrolment resolves student and course membership. Each submission belongs to one student and assignment. An unsuccessful submission has no receipt; an accepted submission must have exactly one. Enforcing submission identifier uniqueness prevents duplicate confirmation when a client retries. File storage and metadata must agree before acceptance, with recovery handling partial failures. These rules support FR3, FR7 and the integration tests.

5. Team roles and project planning

5.1 Responsibilities

The Product Owner sets priorities and accepts business outcomes. The Scrum Master facilitates events and removes impediments. Developers design, implement and integrate the LMS. A quality engineer develops tests and reports defects; a security and accessibility specialist reviews protection and inclusive interaction. Specialists contribute within the Scrum Developers accountability rather than forming a separate approval hierarchy.

5.2 Indicative delivery timeline

Table 3. Proposed lifecycle timeline

Weeks	Delivery focus and milestone
1 to 2	Elicitation, policy clarification and initial backlog.
3 to 6	Identity, enrolment and materials increments.
7 to 12	Submissions, receipts and deadline dashboard.
13 to 16	Grading, feedback and course communication.
17 to 20	Integrated security, load and recovery validation.
21 to 24	User pilot, training, migration rehearsal and release.

Dates depend on approval of the project start. Reviews and regression testing occur throughout, including early Sprints. Release requires resolved critical defects, stakeholder acceptance and a successful restoration exercise. After release, the team monitors service, fixes defects and manages changes.

6. Software quality and testing

6.1 Assignment submission test plan

Prepare an enrolled student, a nonmember, a lecturer, open and expired assignments, and safe test files. Unit tests check validation and deadline logic; integration tests check storage and database consistency; system tests exercise the browser workflow. User acceptance testing confirms the receipt and lecturer view. OWASP (n.d.) informs upload abuse cases. These are planned tests, not executed results.

Table 4. Submission test cases and expected outcomes

Test and level	Input or action	Expected outcome
T1 Unit	Valid PDF before deadline.	Validation succeeds.
T2 Unit	Completion after deadline, no extension.	Rejected as late.
T3 Integration	Storage fails before acceptance.	No success receipt; incomplete record cleaned up.
T4 Integration	Retry the same request identifier.	Same receipt; no duplicate submission.
T5 System	Nonmember attempts upload or download.	Access denied; no private file exposed.
T6 System	Disallowed type or file above 20 MB.	Safe rejection with clear explanation.
T7 Acceptance	Student submits; lecturer opens it.	Correct file, timestamp and receipt agree.
T8 System	Compare assignment views.	Deadline and submission status agree.

Entry requires stable test data and agreed criteria. Exit requires all critical cases passing and no unresolved critical defects. Record evidence, severity, owner and retest outcome; load and accessibility checks additionally verify NFR2 and NFR5.

7. Risk management

7.1 Assessment matrix

Likelihood and impact each range from 1 to 5. Their product determines low (1 to 7), medium (8 to 14) or high (15 to 25) priority. Initial scores are estimates requiring review each Sprint.

Likelihood	Impact 1	Impact 2	Impact 3	Impact 4	Impact 5
5	5	10	15	20	25
4	4	8	12	R3, R4	20
3	3	6	9	R5	R1, R2
2	2	4	6	8	10
1	1	2	3	4	5

Low: 1 to 7 · Medium: 8 to 14 · High: 15 to 25

Figure 3. Likelihood and impact risk matrix

Table 5. Risk responses and owners

Risk	Score	Mitigation and owner
R1 Data exposure	3 × 5 = 15	Security lead: permission tests, restricted storage and incident response.
R2 Submission loss	3 × 5 = 15	Developer: durable receipts, retries and recovery rehearsal.
R3 Deadline traffic	4 × 4 = 16	Operations lead: load tests, monitoring and capacity reserve.
R4 Scope growth	4 × 4 = 16	Product Owner: prioritise essentials and defer optional export.
R5 Excluded users	3 × 4 = 12	Accessibility specialist: assistive technology and user testing.

Escalate unresolved high risks before release. Keep rollback procedures and a support channel ready during the pilot.

8. Ethical and legal considerations

8.1 Privacy and security

Submissions and grades contain personal information. Malaysia's Personal Data Protection Act and its 2024 amendments inform privacy governance (Personal Data Protection Department, n.d.). Collect only necessary information, explain its purpose, restrict access and define retention periods. The university must assess applicable breach notification and governance obligations. Longer sessions should preserve convenience without exposing records on shared devices.

8.2 Copyright

Uploading a textbook does not establish permission to distribute it. The Copyright Act 1987 is relevant to course materials (Intellectual Property Corporation of Malaysia, n.d.). Record licences, attribute sources, restrict access appropriately and provide a takedown procedure. Educational use should not be treated as unrestricted permission to reproduce works.

8.3 Accessibility and academic fairness

Inaccessible controls can prevent timely submissions. Apply WCAG 2.2 AA, provide meaningful labels and textual status messages, and test keyboard access and reflow (W3C, 2024). Verify problems with affected users rather than assuming every reported accessibility defect exists locally.

Submission receipts support fair dispute resolution but cannot prove authorship. Preserve audit records and allow human review of exceptional cases. Automated alerts must not independently determine misconduct or academic penalties.

8.4 Recommendation

Proceed with the proposed LMS through incremental validation. Prioritise reliable submission and protected access, confirm local user needs, and use the pilot to decide whether the release criteria are met.

References

Intellectual Property Corporation of Malaysia (n.d.) *Copyright Act*. Available at: https://www.myipo.gov.my/copyright-act/ (Accessed: 5 November 2026).

Kiruba, S. (2026) 'Review of MyCamu Lite', *Google Play*, 3 August. Available at: https://play.google.com/store/apps/details?id=com.octoze.camu.mycamuapp (Accessed: 5 November 2026).

OWASP (n.d.) *File Upload Cheat Sheet*. Available at: https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html (Accessed: 5 November 2026).

Personal Data Protection Department (n.d.) *Akta Perlindungan Data Peribadi (Pindaan) 2024*. Available at: https://www.pdp.gov.my/akta/akta-pdppindaan-2024/ (Accessed: 5 November 2026).

Punay, A.R. (2026) 'Submitting Outputs', *Mapúa Malayan Digital College Support*, updated 19 June. Available at: https://support.mmdc.mcl.edu.ph/hc/en-us/articles/17566038499341-Submitting-Outputs (Accessed: 5 November 2026).

Schwaber, K. and Sutherland, J. (2020) *The Scrum Guide*. November. Available at: https://scrumguides.org/scrum-guide.html (Accessed: 5 November 2026).

Suriyakumar, P. (2021) 'Review of MyCamu Lite', *Google Play*, 10 January. Available at: https://play.google.com/store/apps/details?id=com.octoze.camu.mycamuapp (Accessed: 5 November 2026).

W3C (2024) *Web Content Accessibility Guidelines (WCAG) 2.2*. Recommendation, 12 December. Available at: https://www.w3.org/TR/WCAG22/ (Accessed: 5 November 2026).

Appendix A. Student feedback

A.1 Method and questions

Four students provided feedback on 4 November 2026 about their own Camu accounts and agreed to anonymous use. They rated ease of finding materials and deadlines, described observed problems and access to messages, grades and feedback, suggested three changes, and identified useful features. Their answers are summarised below.

A.2 Anonymised responses

Participant	Rating	Summary
Student 1	2/5	Tasks showed status; Content and Assignments were empty. Messages failed. Requested materials, consistent assignments and update alerts. Valued class times and task status.
Student 2	3/5	Tasks showed statuses but Assignments was empty. Messages failed and Gradebook showed no coursework. Requested consistent lists, visible deadlines and task linked feedback. Valued status labels.
Student 3	2/5	Content and Assignments were empty; Messages failed. Requested course materials, reliable messages, consistent deadlines and task linked feedback. Valued the calendar.
Student 4	3/5	Tasks contained entries while Assignments was empty; Messages failed. Requested reliable messages, consistent task information, materials and informative empty states. Valued Calendar and Tasks.